

USB PC UPGRADE

1. Blank Flash Upgrade

1.1. Usage Scenarios And Principles

There is nothing in norflash or nandflash, please use pcb's usb to connect pc, download uboot to memory, execute uboot and pc software to interact to complete the download and upgrade.

1.2. Requirement

u-boot.bin for booting, image and pc tool for upgrade.

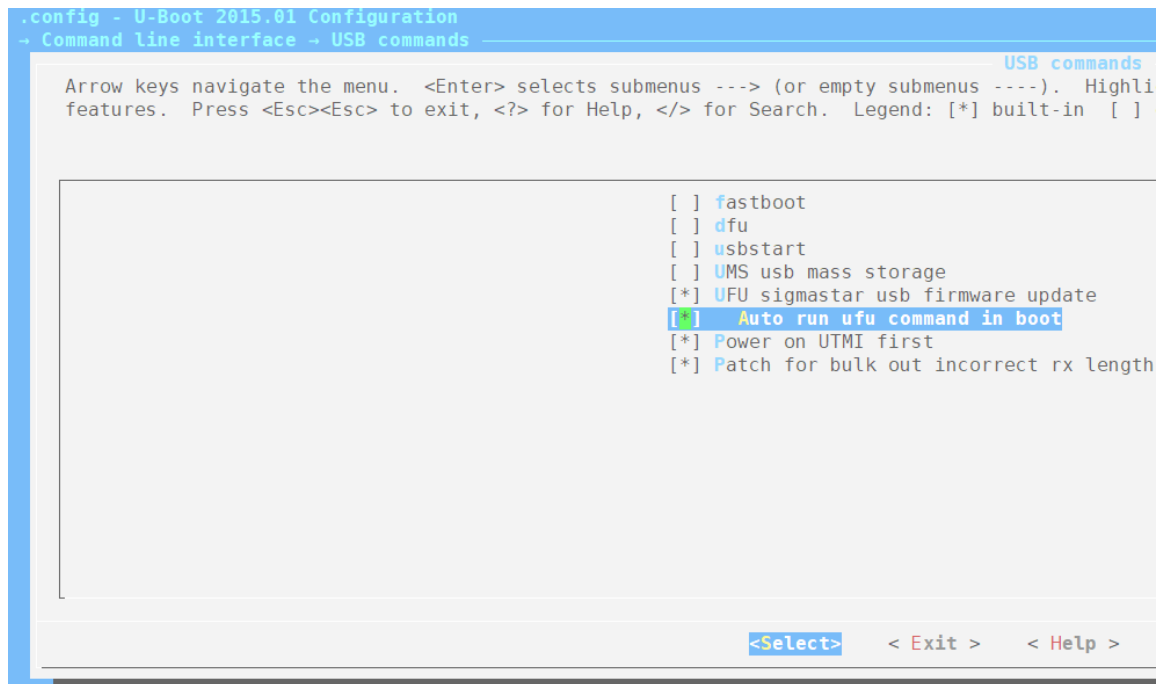
1.3. Image

Refer to [Environment setup](#)

[../..../..../Environmentsetup/Environmentsetup.html] to build `image :`
`project/image/output/images` .

1.4. BOOT

Enable the following options.



Rename the generated `u-boot_spinand.img.bin` or `u-boot.img.bin` to `u-boot.bin` (only for blank boot). This file will be used in the following steps.

1.5. Pc Tool

[Click to download PC Tool](#) [media/usbdownloadtool.rar].

1.6. Steps

Take norflash as an example:

1. release pc tool to images

usb-device升级 > P3_demo_norflash > images

名称	修改日期	类型	大小
boot	2020/10/21 15:21	文件夹	
scripts	2020/10/21 15:21	文件夹	
AitUVCExtApi.dll	2020/9/29 22:04	应用程序扩展	182 KB
appconfigs.jffs2	2020/10/21 15:21	JFFS2 文件	832 KB
auto_update.txt	2020/10/21 15:21	TXT 文件	1 KB
customer.jffs2	2020/10/21 15:21	JFFS2 文件	5,504 KB
ipl_cust_s.bin	2020/10/21 15:21	BIN 文件	26 KB
ipl_s.bin	2020/10/21 15:21	BIN 文件	20 KB
kernel	2020/10/21 15:21	文件	1,838 KB
miservice.sqfs	2020/10/21 15:21	SQFS 文件	4,916 KB
partition_layout.txt	2020/10/21 15:21	TXT 文件	1 KB
rootfs.sqfs	2020/10/21 15:21	SQFS 文件	1,724 KB
u-boot.bin	2020/10/12 16:11	BIN 文件	531 KB
uboot_s.bin	2020/10/21 15:21	BIN 文件	93 KB
usb_updater.bin	2020/10/13 12:09	BIN 文件	64 KB
USBDownloadTool-nopad.exe	2020/9/29 22:04	应用程序	2,251 KB

2. release u-boot.bin

Replace the booting **u-boot.bin** here.

usb-device升级 > P3_demo_norflash > images

名称	修改日期	类型	大小
boot	2020/10/21 15:21	文件夹	
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partition_layout.txt	2020/10/21 15:21	TXT 文件	1 KB
rootfs.sqfs	2020/10/21 15:21	SQFS 文件	1,724 KB
u-boot.bin	2020/10/21 14:24	BIN 文件	277 KB
u-boot.bin.old	2020/10/12 16:11	OLD 文件	531 KB
u-boot.img.bin	2020/10/21 14:24	BIN 文件	277 KB
uboot_s.bin	2020/10/21 15:21	BIN 文件	93 KB
usb_updater.bin	2020/10/13 12:09	BIN 文件	64 KB
USBDownloadTool-nopad.exe	2020/9/29 22:04	应用程序	2,251 KB

3. Some versions need to add images upgrade script (please ignore this step, if you use nandflash)

Add the following script in **images\scripts\[m]xp**.

```
sf probe 0

mxp r.info MXPT

sf erase $(sf_part_start) $(sf_part_size)

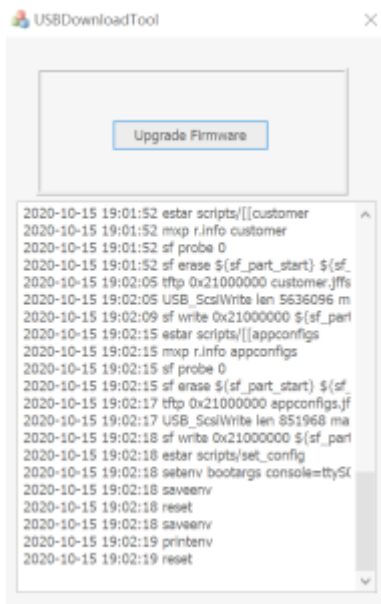
sf write 0x21000000 $(sf_part_start) $(filesize)
```

```
1 # <- this is for comment / total file size must be less than 4KB
2 mxp t.init
3 tftp 0x21000000 boot/MXP_SF.bin
4 mxp t.update 0x21000000
5 mxp t.load
6 sf probe 0
7 mxp r.info MXPT
8 sf erase $(sf_part_start) $(sf_part_size)
9 sf write 0x21000000 $(sf_part_start) $(filesize)
10 % <- this is end of file symbol
11 *
```

If there are similar codes as follows in the version, you don't need to add it.

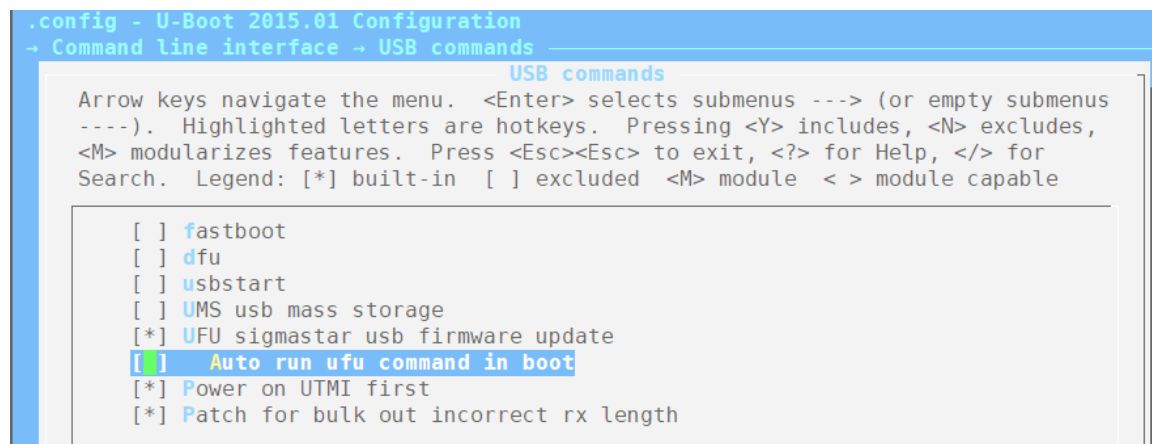
```
1 # <- this is for comment / total file size must be less than 4KB
2 mxp t.init
3 tftp 0x21000000 boot/MXP_SF.bin
4 mxp t.update 0x21000000
5 mxp t.load
6 mxp r.info MXPT
7 mxp t.init
8 mxp t.update 0x21000000
9 mxp t.load
10 % <- this is end of file symbol
11
```

4. Connect to the pc by usb, power on, and execute `USBDownloadTool-nopad.exe`.



2. USB-PC Upgrade In UBOOT

Norflash or nandflash already has uboot, and UBOOT supports PC to upgrade in device mode, execute uboot and pc software to interact to complete the download and upgrade.



This uboot supports ufu upgrade by default, but it does not run automatically.

Set the following parameters in boot, and you can enter the device upgrade mode after booting.

```
setenv ota_upgrade_status 1
```

saveenv